Preliminary results

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Nonlinear Exposure–Response Analysis (Xinyin Miao)

Load packages

```
library(dplyr)
library(ggplot2)
library(splines)
library(lubridate)
library(purrr)
library(tidyr)
```

Read and prepare data

```
df <- read.csv("data/analysis_2024.csv") |>
  mutate(
    date = as.Date(date),
    time = seq_len(n()),
    month_name = factor(month(date, label = TRUE, abbr = TRUE),
                        levels = month.abb),
    season = case_when(
      month(date) %in% c(12, 1, 2) ~ "Winter",
      month(date) %in% c(3, 4, 5)  ~ "Spring",
      month(date) %in% c(6, 7, 8)  ~ "Summer",
      TRUE                         ~ "Fall"
    ) |> factor(levels = c("Winter", "Spring", "Summer", "Fall")),
    weekday_name = factor(
      weekdays(date),
      levels = c("Monday", "Tuesday", "Wednesday",
                 "Thursday", "Friday", "Saturday", "Sunday")
    ),
    is_weekend = as.numeric(is_weekend)
  )

str(df)
```

```
## 'data.frame':    124 obs. of  18 variables:
##  $ date         : Date, format: "2024-01-01" "2024-01-04" ...
##  $ pm25         : num  8.67 6.78 2.69 2.65 3.48 ...
##  $ mental_health: int  449 505 395 494 433 463 530 517 533 434 ...
##  $ suicide      : int  7 5 2 9 3 3 4 4 10 6 ...
##  $ tmax         : num  7.4 6.7 3 13.8 14.4 -0.8 -0.9 3.7 10.6 3.8 ...
##  $ tmin         : num  -0.4 -3 -0.1 4.8 1.2 -6.2 -4.3 -6.6 6.8 2.2 ...
##  $ tmean        : num  3.3 1.9 1.5 8.8 8.8 -2.9 -2.4 -2.1 8.6 3.2 ...
```

```
##  $ precip       : num  0.1 0.3 6.8 21 19.5 9.1 2.5 0 6.1 21.1 ...
##  $ windspeed    : num  14.7 25.8 25.2 41.3 34.9 24.8 17.8 18.8 21.8 29 ...
##  $ rh_max       : int  91 85 100 92 96 94 79 74 100 97 ...
##  $ month        : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ weekday      : int  0 3 6 2 5 1 4 0 3 6 ...
##  $ is_weekend   : num  0 0 1 0 1 0 0 0 0 1 ...
##  $ heat_day     : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ time         : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ month_name   : Ord.factor w/ 12 levels "Jan"<"Feb"<"Mar"<..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ season       : Factor w/ 4 levels "Winter","Spring",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ weekday_name : Factor w/ 7 levels "Monday","Tuesday",..: 1 4 7 3 6 2 5 1 4 7 ...
```

```r
summary(df[, c("pm25", "mental_health", "suicide", "tmean", "precip", "is_weekend")])
```

```
##       pm25        mental_health      suicide          tmean
##  Min.   : 1.833   Min.   :368.0   Min.   : 1.00   Min.   :-9.500
##  1st Qu.: 4.473   1st Qu.:445.5   1st Qu.: 4.00   1st Qu.: 6.575
##  Median : 5.662   Median :496.0   Median : 5.00   Median :14.800
##  Mean   : 6.756   Mean   :485.5   Mean   : 5.29   Mean   :13.326
##  3rd Qu.: 8.665   3rd Qu.:529.2   3rd Qu.: 7.00   3rd Qu.:20.425
##  Max.   :23.188   Max.   :601.0   Max.   :14.00   Max.   :29.200
##      precip         is_weekend
##  Min.   : 0.000   Min.   :0.0000
##  1st Qu.: 0.000   1st Qu.:0.0000
##  Median : 0.100   Median :0.0000
##  Mean   : 3.571   Mean   :0.2903
##  3rd Qu.: 2.550   3rd Qu.:1.0000
##  Max.   :39.400   Max.   :1.0000
```

Model specification

Primary nonlinear model:

$$\log(\mu_t) = \beta_0 + \text{ns}(\text{PM}_{2.5,t}, 3) + \text{ns}(\text{time}_t, 4) + \text{ns}(\text{tmean}_t, 3) + \beta_1 \cdot \text{is_weekend}_t + \beta_2 \cdot \text{precip}_t$$

where

$$Y_t \sim \text{Quasi-Poisson}, \qquad \mu_t = E(Y_t).$$

For comparison, the linear specification is:

$$\log(\mu_t) = \beta_0 + \beta_1 \cdot \text{PM}_{2.5,t} + \text{ns}(\text{time}_t, 4) + \text{ns}(\text{tmean}_t, 3) + \beta_2 \cdot \text{is_weekend}_t + \beta_3 \cdot \text{precip}_t$$

Fit linear and nonlinear models

Mental health outcome

```r
fit_mh_linear <- glm(
  mental_health ~ pm25 + ns(time, 4) + ns(tmean, 3) + is_weekend + precip,
  family = quasipoisson(link = "log"),
  data = df
)
```

```r
fit_mh_nonlinear <- glm(
  mental_health ~ ns(pm25, 3) + ns(time, 4) + ns(tmean, 3) + is_weekend + precip,
  family = quasipoisson(link = "log"),
  data = df
)

summary(fit_mh_linear)
```

```
##
## Call:
## glm(formula = mental_health ~ pm25 + ns(time, 4) + ns(tmean,
##     3) + is_weekend + precip, family = quasipoisson(link = "log"),
##     data = df)
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.1079674  0.0481610 126.824  < 2e-16 ***
## pm25          -0.0027390  0.0022091  -1.240 0.217593
## ns(time, 4)1  -0.1885992  0.0767155  -2.458 0.015472 *
## ns(time, 4)2  -0.1272162  0.0481845  -2.640 0.009457 **
## ns(time, 4)3  -0.0699661  0.0716241  -0.977 0.330730
## ns(time, 4)4  -0.0344335  0.0307842  -1.119 0.265705
## ns(tmean, 3)1  0.2279141  0.0480185   4.746 6.11e-06 ***
## ns(tmean, 3)2  0.4927748  0.1061062   4.644 9.28e-06 ***
## ns(tmean, 3)3  0.2499008  0.0727788   3.434 0.000833 ***
## is_weekend    -0.1858636  0.0145385 -12.784  < 2e-16 ***
## precip        -0.0020177  0.0008388  -2.405 0.017775 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 2.230983)
##
##     Null deviance: 727.73  on 123  degrees of freedom
## Residual deviance: 253.91  on 113  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 3
```

```r
summary(fit_mh_nonlinear)
```

```
##
## Call:
## glm(formula = mental_health ~ ns(pm25, 3) + ns(time, 4) + ns(tmean,
##     3) + is_weekend + precip, family = quasipoisson(link = "log"),
##     data = df)
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.1255768  0.0588560 104.077  < 2e-16 ***
## ns(pm25, 3)1   0.0291699  0.0356936   0.817  0.41555
## ns(pm25, 3)2  -0.1361293  0.0711396  -1.914  0.05825 .
## ns(pm25, 3)3  -0.1455901  0.0633298  -2.299  0.02338 *
## ns(time, 4)1  -0.1538828  0.0781149  -1.970  0.05133 .
## ns(time, 4)2  -0.1062392  0.0495617  -2.144  0.03425 *
```

```
## ns(time, 4)3  -0.0551672  0.0711053  -0.776  0.43949
## ns(time, 4)4  -0.0304625  0.0309187  -0.985  0.32665
## ns(tmean, 3)1  0.2113338  0.0488964   4.322 3.38e-05 ***
## ns(tmean, 3)2  0.4633315  0.1069611   4.332 3.26e-05 ***
## ns(tmean, 3)3  0.2190613  0.0739955   2.960  0.00376 **
## is_weekend    -0.1895067  0.0148372 -12.772  < 2e-16 ***
## precip        -0.0020941  0.0008364  -2.504  0.01375 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 2.179859)
##
##     Null deviance: 727.73  on 123  degrees of freedom
## Residual deviance: 244.17  on 111  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 3
```

Suicide outcome

```r
fit_sui_linear <- glm(
  suicide ~ pm25 + ns(time, 4) + ns(tmean, 3) + is_weekend + precip,
  family = quasipoisson(link = "log"),
  data = df
)

fit_sui_nonlinear <- glm(
  suicide ~ ns(pm25, 3) + ns(time, 4) + ns(tmean, 3) + is_weekend + precip,
  family = quasipoisson(link = "log"),
  data = df
)

summary(fit_sui_linear)
```

```
##
## Call:
## glm(formula = suicide ~ pm25 + ns(time, 4) + ns(tmean, 3) + is_weekend +
##     precip, family = quasipoisson(link = "log"), data = df)
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.736720   0.350293   2.103  0.03767 *
## pm25           0.008381   0.013412   0.625  0.53331
## ns(time, 4)1  -0.308464   0.473974  -0.651  0.51649
## ns(time, 4)2  -0.256250   0.300188  -0.854  0.39511
## ns(time, 4)3  -0.424342   0.463240  -0.916  0.36160
## ns(time, 4)4  -0.161496   0.204770  -0.789  0.43196
## ns(tmean, 3)1  0.974544   0.314486   3.099  0.00245 **
## ns(tmean, 3)2  2.228649   0.779844   2.858  0.00508 **
## ns(tmean, 3)3  0.519953   0.453429   1.147  0.25392
## is_weekend    -0.031001   0.088285  -0.351  0.72613
## precip        -0.004389   0.005359  -0.819  0.41450
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## (Dispersion parameter for quasipoisson family taken to be 0.9540781)
##
##     Null deviance: 129.88  on 123  degrees of freedom
## Residual deviance: 108.10  on 113  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```

```r
summary(fit_sui_nonlinear)
```

```
##
## Call:
## glm(formula = suicide ~ ns(pm25, 3) + ns(time, 4) + ns(tmean,
##     3) + is_weekend + precip, family = quasipoisson(link = "log"),
##     data = df)
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.145531   0.415638   2.756  0.00684 **
## ns(pm25, 3)1  -0.012199   0.218571  -0.056  0.95559
## ns(pm25, 3)2  -0.369700   0.427636  -0.865  0.38917
## ns(pm25, 3)3   0.158129   0.354445   0.446  0.65637
## ns(time, 4)1  -0.350083   0.487845  -0.718  0.47451
## ns(time, 4)2  -0.327219   0.310337  -1.054  0.29399
## ns(time, 4)3  -0.393869   0.463369  -0.850  0.39715
## ns(time, 4)4  -0.095809   0.209856  -0.457  0.64889
## ns(tmean, 3)1  1.033787   0.321506   3.215  0.00171 **
## ns(tmean, 3)2  1.970187   0.789624   2.495  0.01406 *
## ns(tmean, 3)3  0.572451   0.467593   1.224  0.22345
## is_weekend    -0.066440   0.091470  -0.726  0.46915
## precip        -0.005562   0.005445  -1.022  0.30921
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 0.9532159)
##
##     Null deviance: 129.88  on 123  degrees of freedom
## Residual deviance: 105.58  on 111  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```

Model comparison

```r
model_compare <- data.frame(
  Model = c("MH linear", "MH nonlinear", "Suicide linear", "Suicide nonlinear"),
  Residual_Deviance = c(
    fit_mh_linear$deviance,
    fit_mh_nonlinear$deviance,
    fit_sui_linear$deviance,
    fit_sui_nonlinear$deviance
  ),
  DF_Residual = c(
```

```r
    fit_mh_linear$df.residual,
    fit_mh_nonlinear$df.residual,
    fit_sui_linear$df.residual,
    fit_sui_nonlinear$df.residual
  ),
  Dispersion = c(
    summary(fit_mh_linear)$dispersion,
    summary(fit_mh_nonlinear)$dispersion,
    summary(fit_sui_linear)$dispersion,
    summary(fit_sui_nonlinear)$dispersion
  )
)

model_compare
```

```
##               Model Residual_Deviance DF_Residual Dispersion
## 1         MH linear          253.9137         113  2.2309826
## 2      MH nonlinear          244.1738         111  2.1798593
## 3    Suicide linear          108.1019         113  0.9540781
## 4 Suicide nonlinear          105.5798         111  0.9532159
```

```r
drop1(fit_mh_nonlinear, test = "F")
```

```
## Single term deletions
##
## Model:
## mental_health ~ ns(pm25, 3) + ns(time, 4) + ns(tmean, 3) + is_weekend +
##     precip
##              Df Deviance  F value    Pr(>F)
## <none>            244.17
## ns(pm25, 3)   3   257.36   1.9974   0.11852
## ns(time, 4)   4   262.11   2.0386   0.09384 .
## ns(tmean, 3)  3   297.89   8.1394 6.018e-05 ***
## is_weekend    1   608.67 165.6982 < 2.2e-16 ***
## precip        1   258.01   6.2891   0.01359 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
drop1(fit_sui_nonlinear, test = "F")
```

```
## Single term deletions
##
## Model:
## suicide ~ ns(pm25, 3) + ns(time, 4) + ns(tmean, 3) + is_weekend +
##     precip
##              Df Deviance F value   Pr(>F)
## <none>            105.58
## ns(pm25, 3)   3   108.47  1.0130 0.389823
## ns(time, 4)   4   107.45  0.4928 0.741031
## ns(tmean, 3)  3   118.37  4.4823 0.005212 **
## is_weekend    1   106.09  0.5328 0.466964
## precip        1   106.61  1.0816 0.300604
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Rate ratio per 10 g/m^3 for linear model

```
get_rr10 <- function(model, var = "pm25") {
  beta <- coef(model)[var]
  se   <- sqrt(vcov(model)[var, var])

  rr    <- exp(beta * 10)
  lower <- exp((beta - 1.96 * se) * 10)
  upper <- exp((beta + 1.96 * se) * 10)

  data.frame(
    term = var,
    beta = beta,
    RR_10 = rr,
    lower_95 = lower,
    upper_95 = upper
  )
}

rr_mh_linear  <- get_rr10(fit_mh_linear, "pm25")
rr_sui_linear <- get_rr10(fit_sui_linear, "pm25")

rr_mh_linear
```

```
##      term         beta     RR_10  lower_95 upper_95
## pm25 pm25 -0.002738997 0.9729817 0.9317521 1.016036
```

```
rr_sui_linear
```

```
##      term        beta    RR_10  lower_95 upper_95
## pm25 pm25 0.008380899 1.087421 0.8360522 1.414367
```

Create exposure–response curve function

```
make_er_curve <- function(model, data, outcome_name,
                          pm_var = "pm25",
                          ref_value = median(data[[pm_var]], na.rm = TRUE),
                          n_points = 200) {

  pm_seq <- seq(min(data[[pm_var]], na.rm = TRUE),
                max(data[[pm_var]], na.rm = TRUE),
                length.out = n_points)

  newdata <- data.frame(
    pm25 = pm_seq,
    time = median(data$time, na.rm = TRUE),
    tmean = median(data$tmean, na.rm = TRUE),
    is_weekend = 0,
    precip = median(data$precip, na.rm = TRUE)
  )

  refdata <- newdata[1, ]
  refdata$pm25 <- ref_value

  X_new <- model.matrix(delete.response(terms(model)), newdata)
```

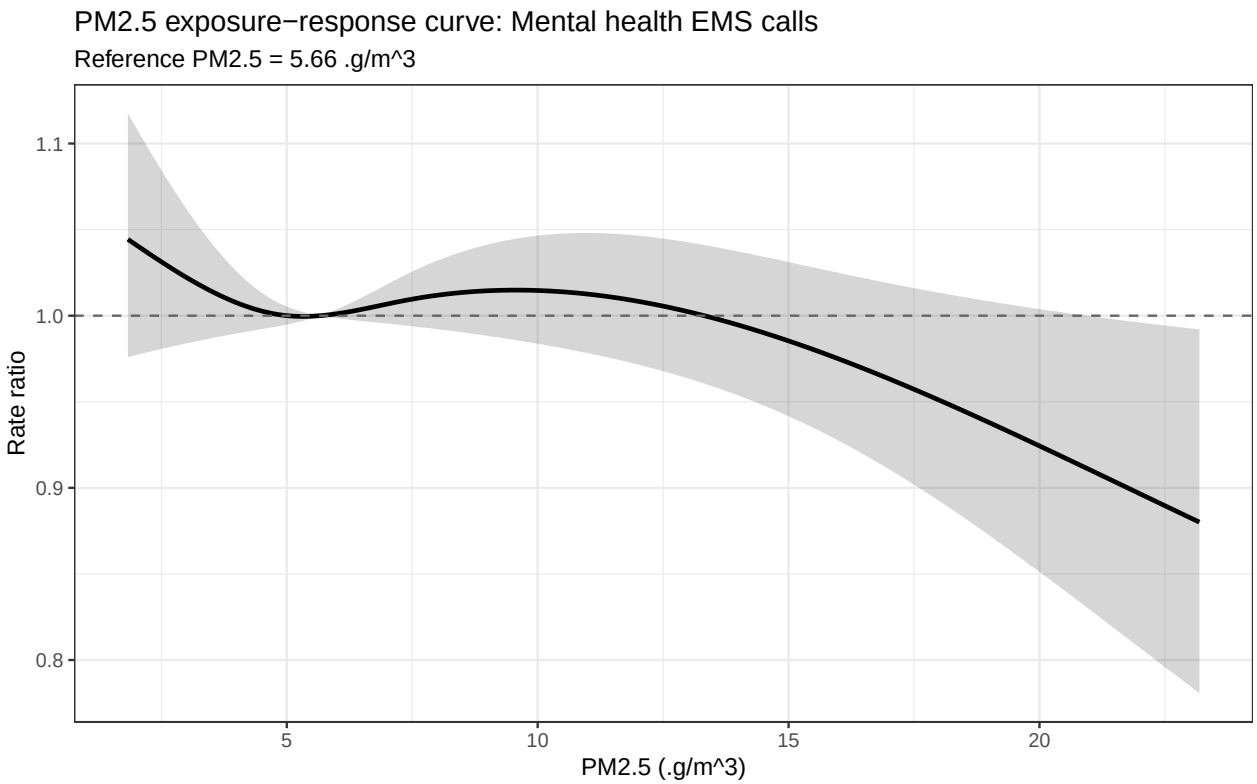
```r
  X_ref <- model.matrix(delete.response(terms(model)), refdata)

  beta <- coef(model)
  V    <- vcov(model)

  diff_mat <- X_new - matrix(X_ref, nrow = nrow(X_new), ncol = ncol(X_new), byrow = TRUE)

  eta_diff <- as.vector(diff_mat %*% beta)
  se_diff  <- sqrt(diag(diff_mat %*% V %*% t(diff_mat)))

  data.frame(
    pm25 = pm_seq,
    RR = exp(eta_diff),
    lower = exp(eta_diff - 1.96 * se_diff),
    upper = exp(eta_diff + 1.96 * se_diff),
    outcome = outcome_name,
    ref_value = ref_value
  )
}

make_linear_curve <- function(model, data, outcome_name,
                              pm_var = "pm25",
                              ref_value = median(data[[pm_var]], na.rm = TRUE),
                              n_points = 200) {

  pm_seq <- seq(min(data[[pm_var]], na.rm = TRUE),
                max(data[[pm_var]], na.rm = TRUE),
                length.out = n_points)

  beta <- coef(model)["pm25"]
  se   <- sqrt(vcov(model)["pm25", "pm25"])

  eta_diff <- beta * (pm_seq - ref_value)
  se_diff  <- abs(pm_seq - ref_value) * se

  data.frame(
    pm25 = pm_seq,
    RR = exp(eta_diff),
    lower = exp(eta_diff - 1.96 * se_diff),
    upper = exp(eta_diff + 1.96 * se_diff),
    outcome = outcome_name,
    model_type = "Linear"
  )
}
```

Generate exposure–response curves

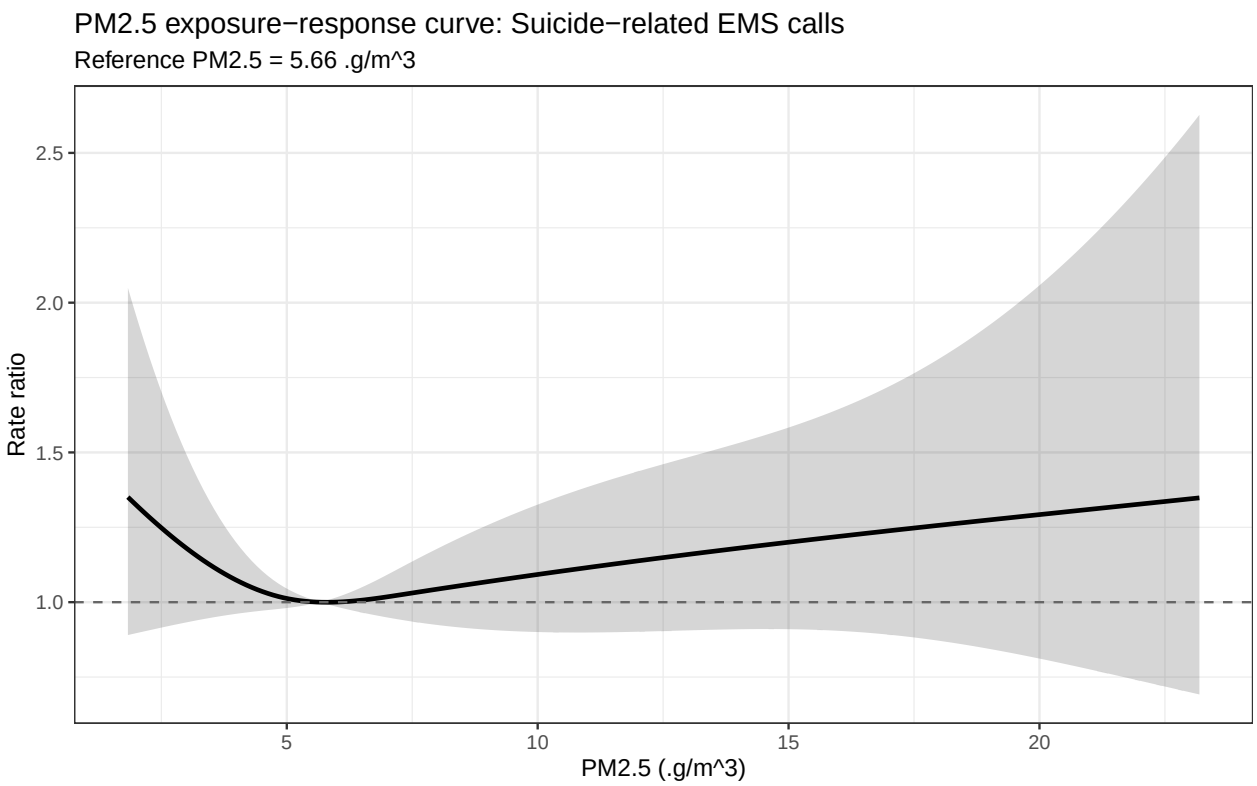
```r
curve_mh  <- make_er_curve(fit_mh_nonlinear, df, "Mental health")
curve_sui <- make_er_curve(fit_sui_nonlinear, df, "Suicide")
```

Plot nonlinear exposure–response curves

```
ggplot(curve_mh, aes(x = pm25, y = RR)) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2, color = "gray40") +
  labs(
    title = "PM2.5 exposure-response curve: Mental health EMS calls",
    subtitle = paste0("Reference PM2.5 = ", round(unique(curve_mh$ref_value), 2), " \u03bcg/m^3"),
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio"
  ) +
  theme_bw()
```



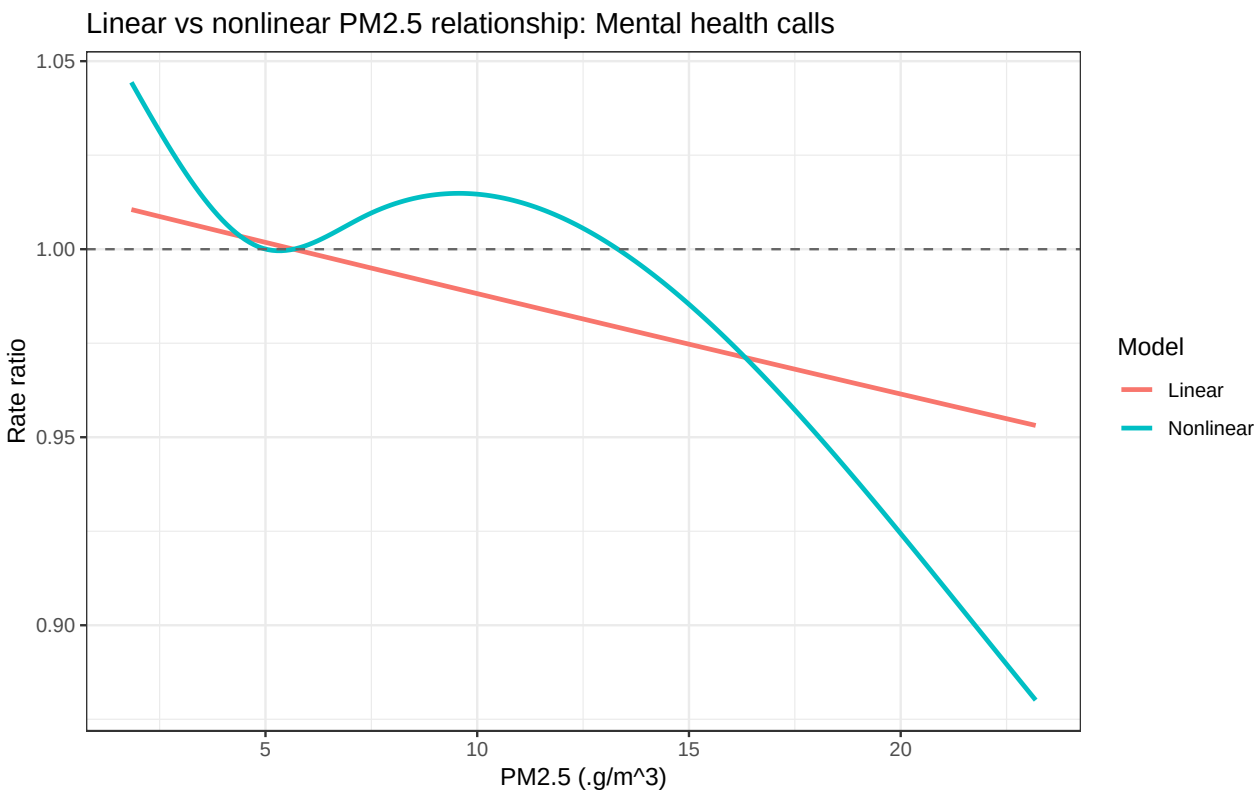

```
ggplot(curve_sui, aes(x = pm25, y = RR)) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2, color = "gray40") +
  labs(
    title = "PM2.5 exposure-response curve: Suicide-related EMS calls",
    subtitle = paste0("Reference PM2.5 = ", round(unique(curve_sui$ref_value), 2), " \u03bcg/m^3"),
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio"
  ) +
  theme_bw()
```



Compare linear vs nonlinear PM2.5 specification

```
curve_mh_linear  <- make_linear_curve(fit_mh_linear, df, "Mental health")
curve_sui_linear <- make_linear_curve(fit_sui_linear, df, "Suicide")

curve_mh_nl_plot  <- curve_mh  |> mutate(model_type = "Nonlinear")
curve_sui_nl_plot <- curve_sui |> mutate(model_type = "Nonlinear")

curve_compare_mh  <- bind_rows(curve_mh_linear, curve_mh_nl_plot)
curve_compare_sui <- bind_rows(curve_sui_linear, curve_sui_nl_plot)
```

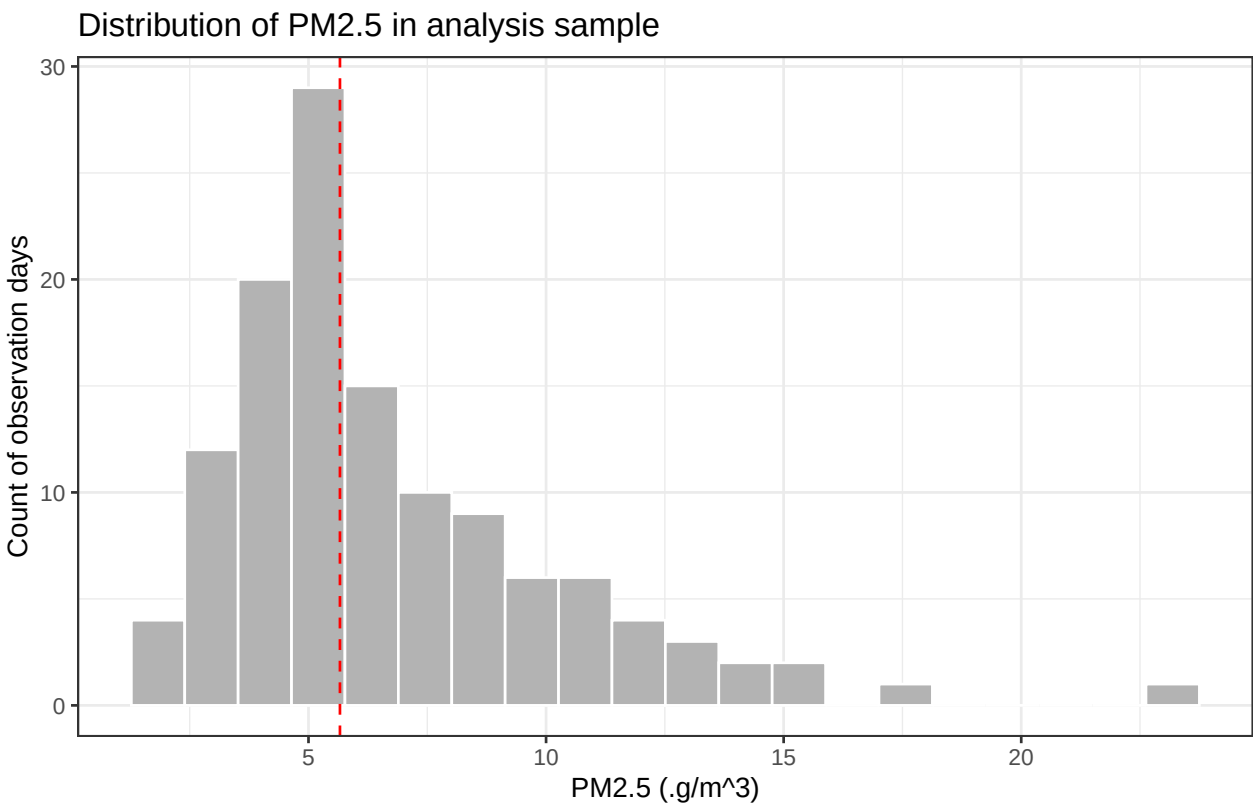
```
ggplot(curve_compare_mh, aes(x = pm25, y = RR, color = model_type)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2, color = "gray40") +
  labs(
    title = "Linear vs nonlinear PM2.5 relationship: Mental health calls",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio",
    color = "Model"
  ) +
  theme_bw()
```




```
ggplot(curve_compare_sui, aes(x = pm25, y = RR, color = model_type)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2, color = "gray40") +
  labs(
    title = "Linear vs nonlinear PM2.5 relationship: Suicide-related calls",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio",
    color = "Model"
  ) +
  theme_bw()
```

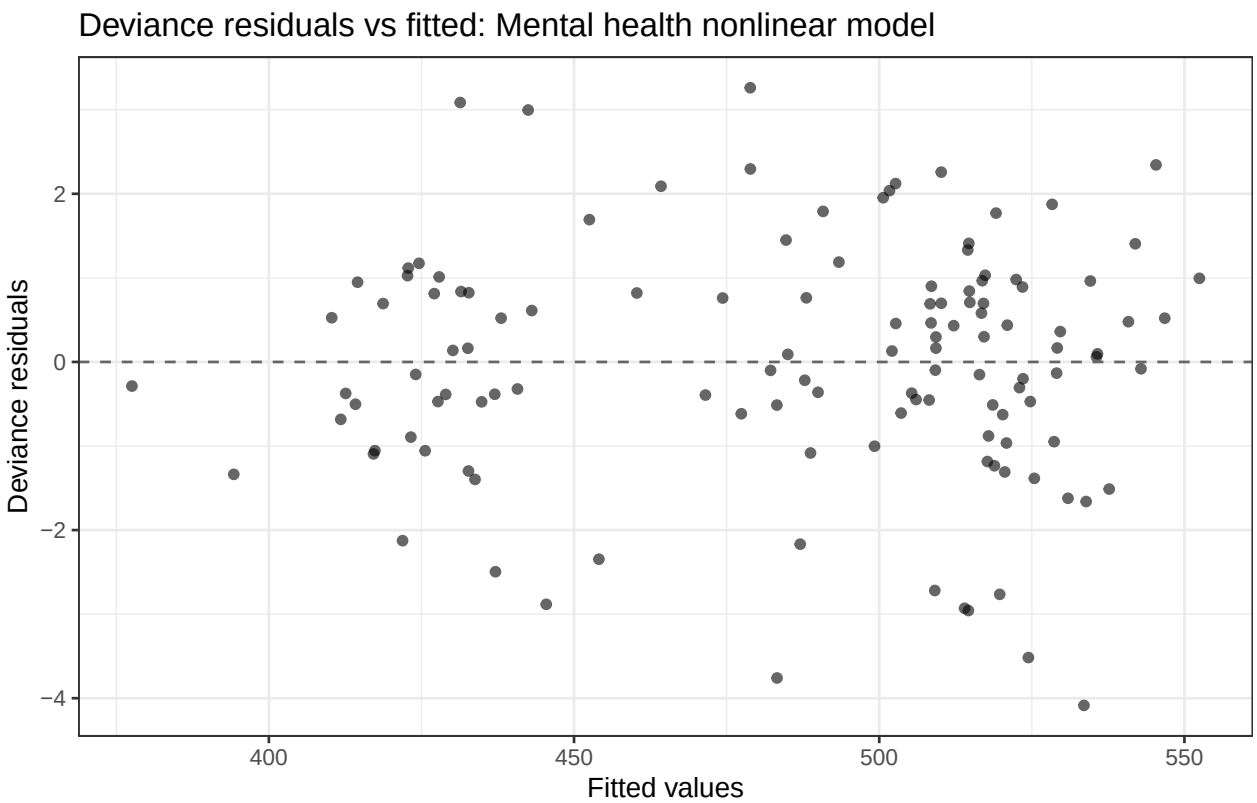
Distribution of PM2.5

```
ggplot(df, aes(x = pm25)) +
  geom_histogram(bins = 20, fill = "gray70", color = "white") +
  geom_vline(xintercept = median(df$pm25, na.rm = TRUE), linetype = 2, color = "red") +
  labs(
    title = "Distribution of PM2.5 in analysis sample",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Count of observation days"
  ) +
  theme_bw()
```

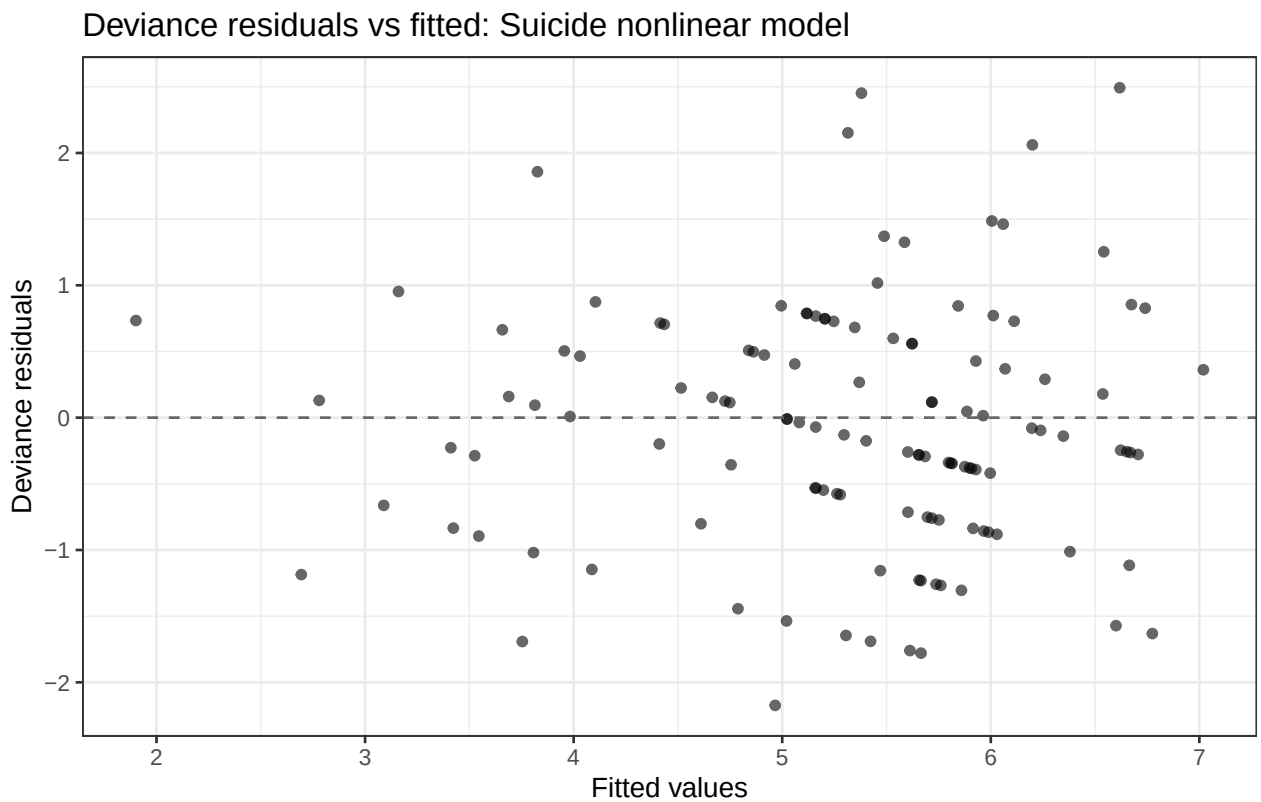


Residual diagnostics

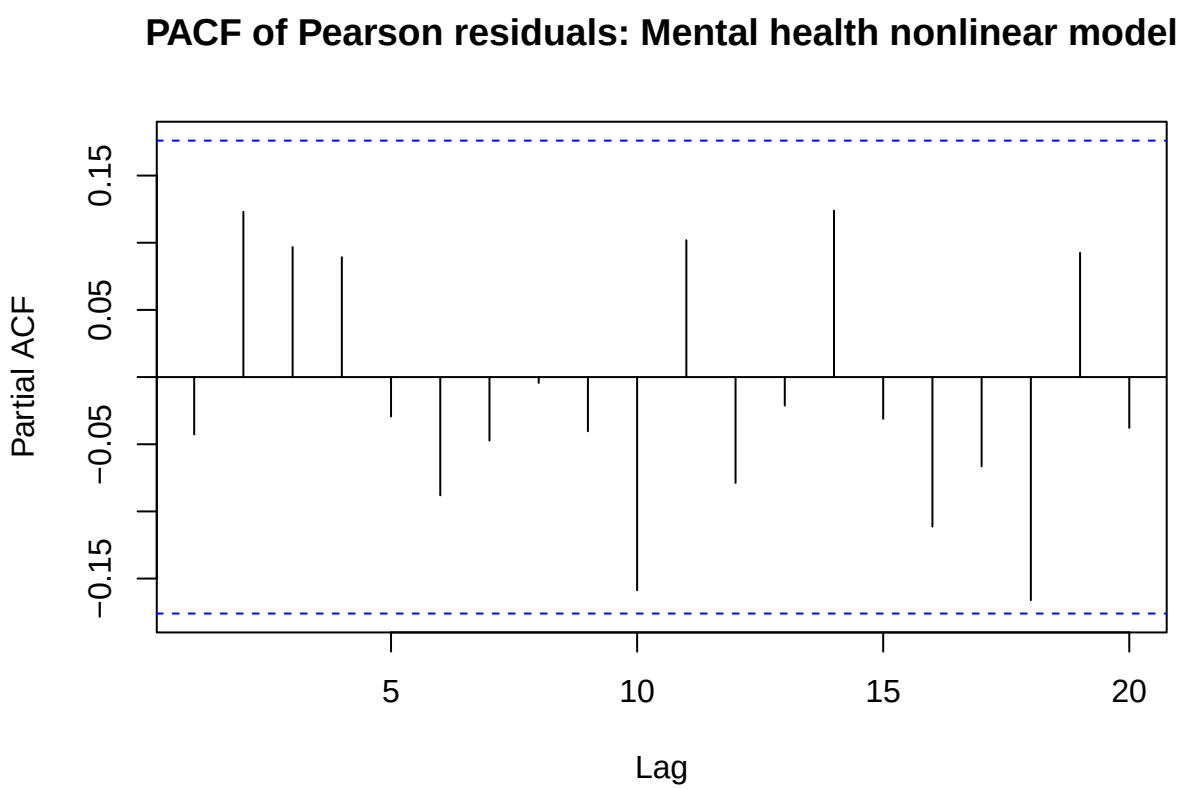
```
diag_mh <- data.frame(
  fitted = fitted(fit_mh_nonlinear),
  resid  = residuals(fit_mh_nonlinear, type = "deviance")
)

ggplot(diag_mh, aes(fitted, resid)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = 2, color = "gray40") +
  labs(
    title = "Deviance residuals vs fitted: Mental health nonlinear model",
    x = "Fitted values",
    y = "Deviance residuals"
  ) +
  theme_bw()
```



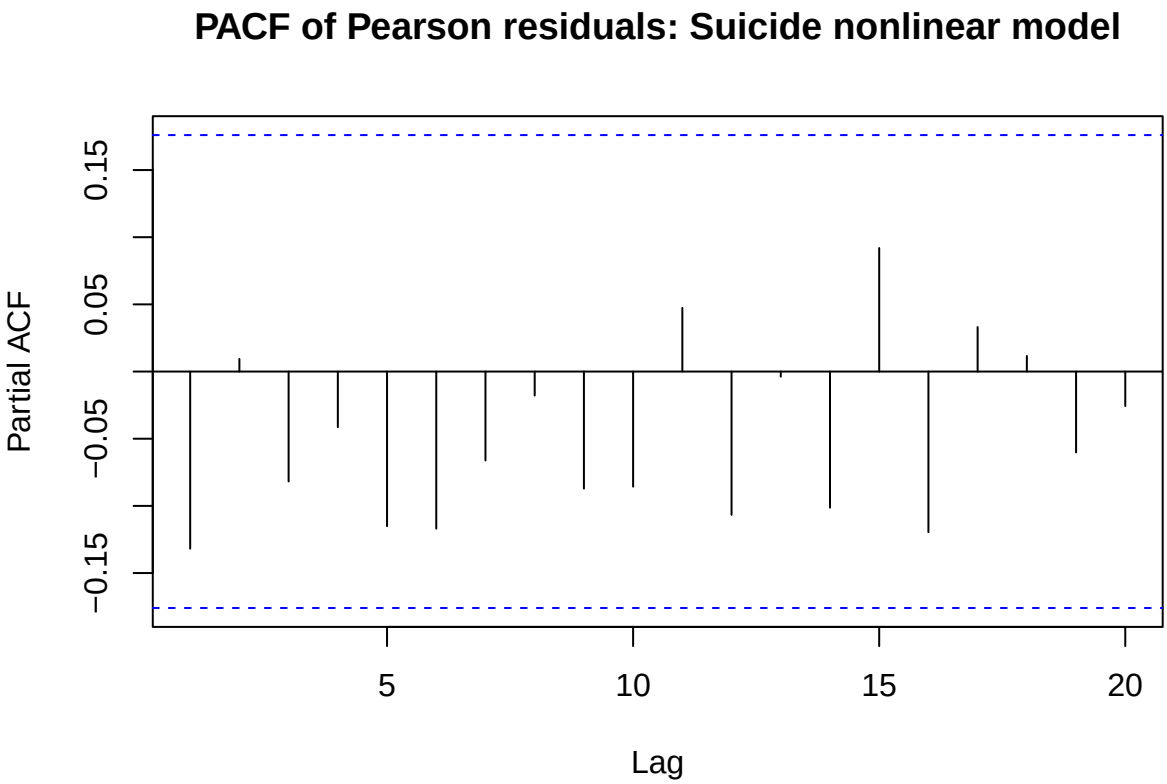

```r
diag_sui <- data.frame(
  fitted = fitted(fit_sui_nonlinear),
  resid  = residuals(fit_sui_nonlinear, type = "deviance")
)

ggplot(diag_sui, aes(fitted, resid)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = 2, color = "gray40") +
  labs(
    title = "Deviance residuals vs fitted: Suicide nonlinear model",
    x = "Fitted values",
    y = "Deviance residuals"
  ) +
  theme_bw()
```




```
pacf(
  residuals(fit_mh_nonlinear, type = "pearson"),
  main = "PACF of Pearson residuals: Mental health nonlinear model"
)
```



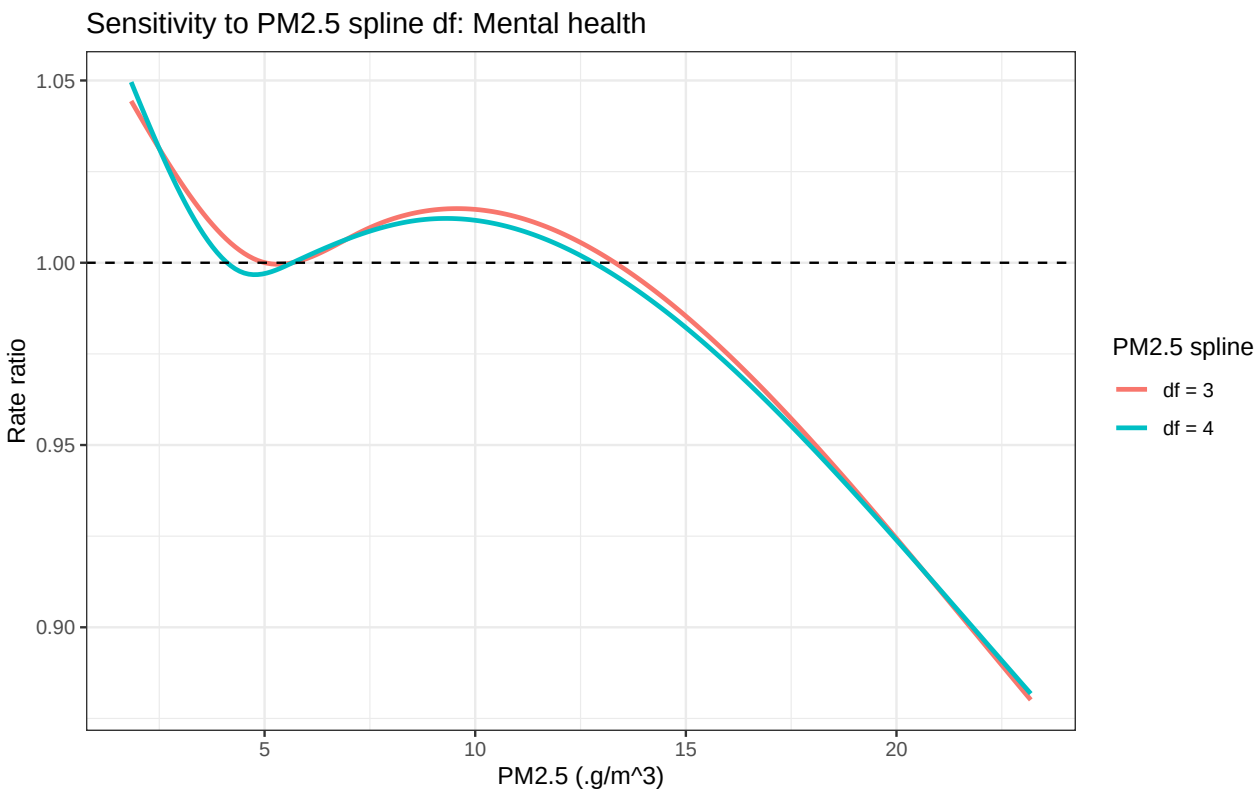
```r
pacf(
  residuals(fit_sui_nonlinear, type = "pearson"),
  main = "PACF of Pearson residuals: Suicide nonlinear model"
)
```



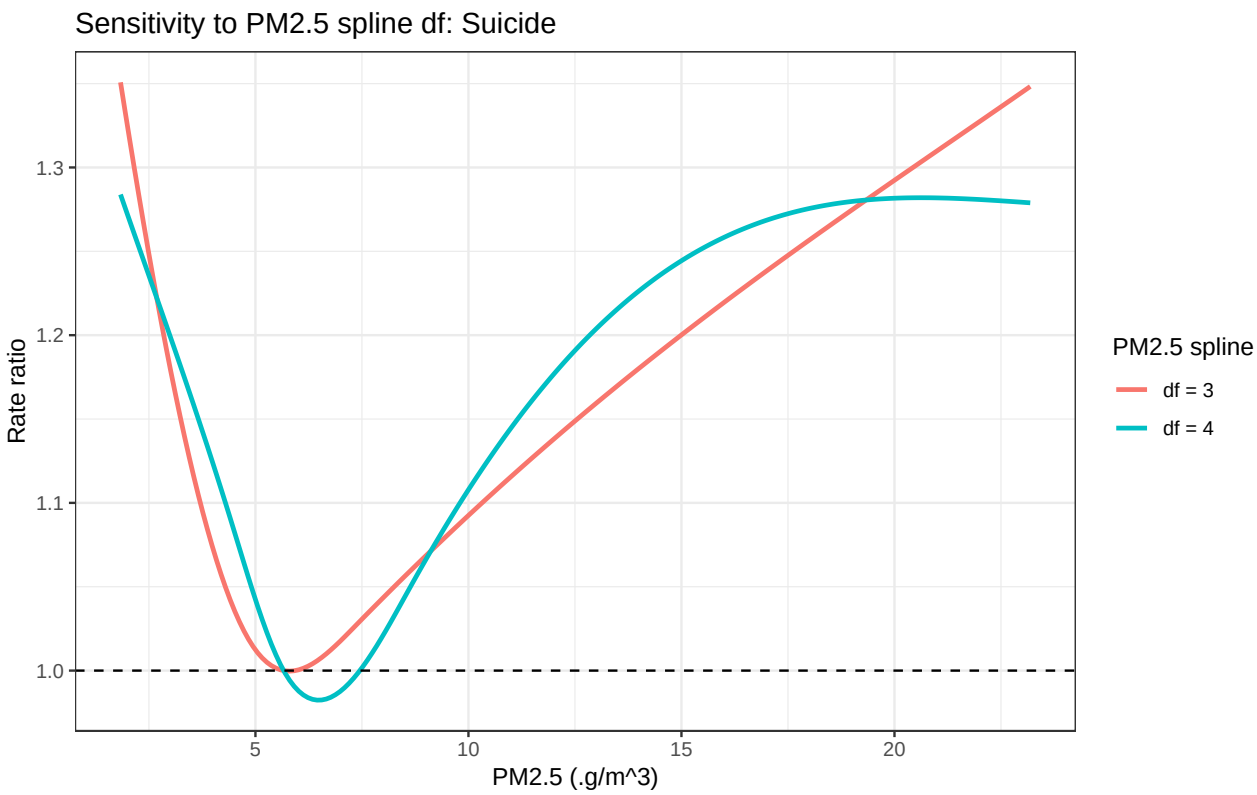
Sensitivity analysis 1: PM2.5 spline df = 4

```r
fit_mh_nonlinear_df4 <- glm(
  mental_health ~ ns(pm25, 4) + ns(time, 4) + ns(tmean, 3) + is_weekend + precip,
  family = quasipoisson(link = "log"),
  data = df
)

fit_sui_nonlinear_df4 <- glm(
  suicide ~ ns(pm25, 4) + ns(time, 4) + ns(tmean, 3) + is_weekend + precip,
  family = quasipoisson(link = "log"),
  data = df
)

curve_mh_df4  <- make_er_curve(fit_mh_nonlinear_df4, df, "Mental health") |> mutate(df_pm = "df = 4")
curve_sui_df4 <- make_er_curve(fit_sui_nonlinear_df4, df, "Suicide") |> mutate(df_pm = "df = 4")

curve_mh_df3  <- curve_mh  |> mutate(df_pm = "df = 3")
curve_sui_df3 <- curve_sui |> mutate(df_pm = "df = 3")

curve_mh_sens  <- bind_rows(curve_mh_df3, curve_mh_df4)
curve_sui_sens <- bind_rows(curve_sui_df3, curve_sui_df4)
```

```r
ggplot(curve_mh_sens, aes(pm25, RR, color = df_pm)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2) +
  labs(
    title = "Sensitivity to PM2.5 spline df: Mental health",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio",
    color = "PM2.5 spline"
  ) +
  theme_bw()
```



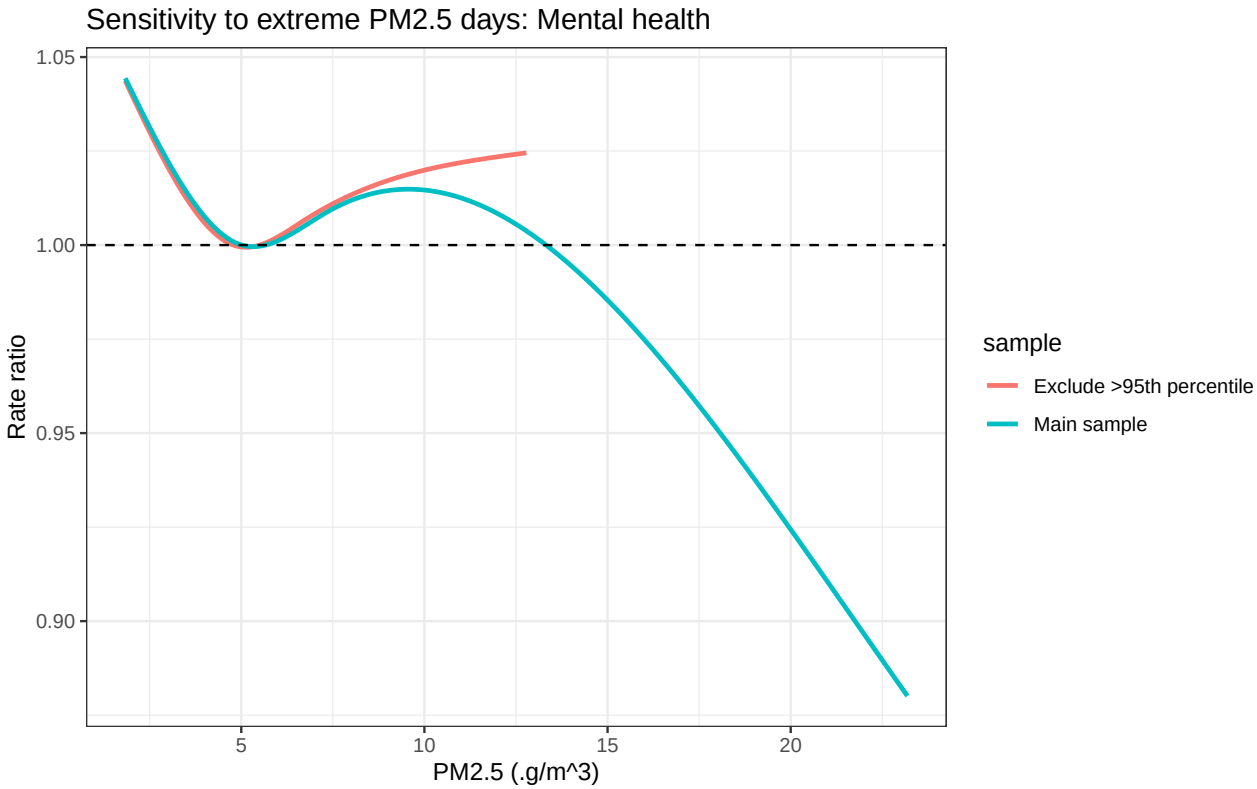

```r
ggplot(curve_sui_sens, aes(pm25, RR, color = df_pm)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2) +
  labs(
    title = "Sensitivity to PM2.5 spline df: Suicide",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio",
    color = "PM2.5 spline"
  ) +
  theme_bw()
```



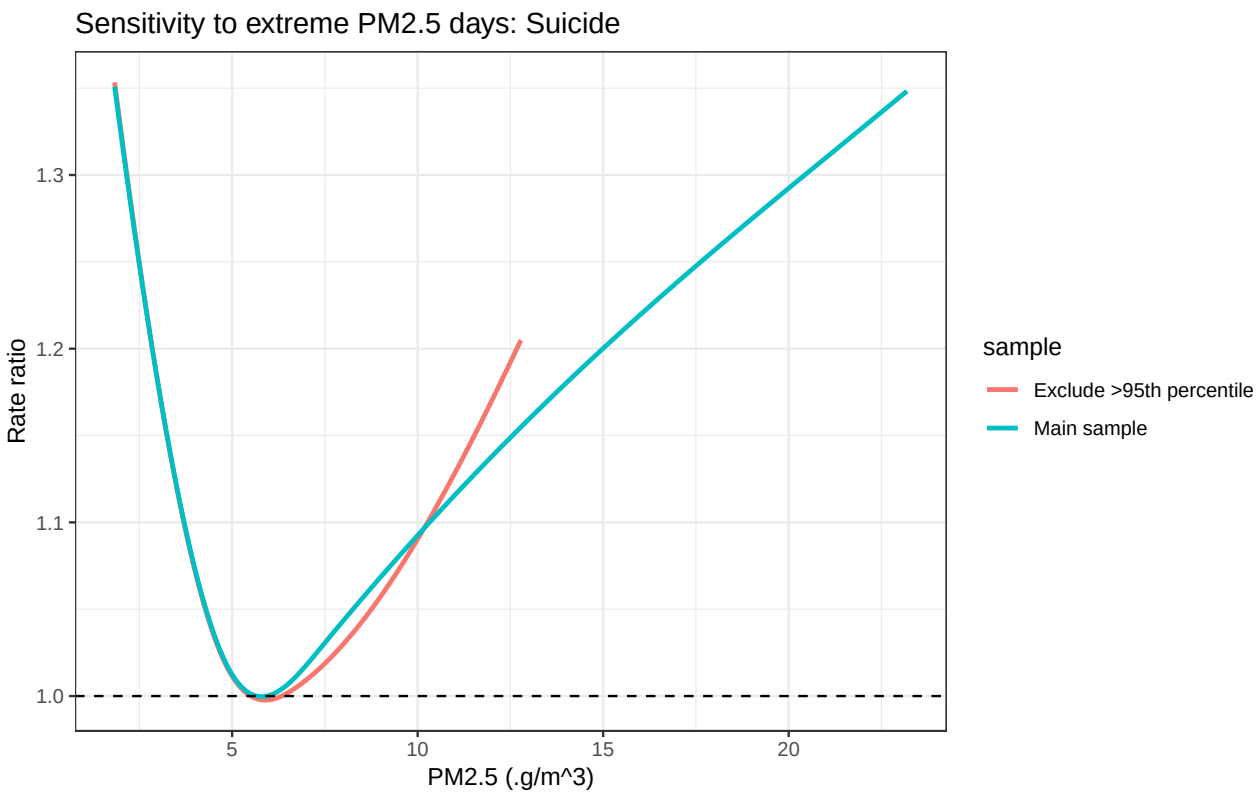
Sensitivity analysis 2: exclude high PM2.5 days

```
pm95 <- quantile(df$pm25, 0.95, na.rm = TRUE)

df_no_high <- df |> filter(pm25 <= pm95)

fit_mh_no_high <- glm(
  mental_health ~ ns(pm25, 3) + ns(time, 4) + ns(tmean, 3) + is_weekend + precip,
  family = quasipoisson(link = "log"),
  data = df_no_high
)

fit_sui_no_high <- glm(
  suicide ~ ns(pm25, 3) + ns(time, 4) + ns(tmean, 3) + is_weekend + precip,
  family = quasipoisson(link = "log"),
  data = df_no_high
)

curve_mh_no_high <- make_er_curve(fit_mh_no_high, df_no_high, "Mental health") |>
  mutate(sample = "Exclude >95th percentile")

curve_sui_no_high <- make_er_curve(fit_sui_no_high, df_no_high, "Suicide") |>
  mutate(sample = "Exclude >95th percentile")

curve_mh_main <- curve_mh |> mutate(sample = "Main sample")
curve_sui_main <- curve_sui |> mutate(sample = "Main sample")
```

```r
ggplot(bind_rows(curve_mh_main, curve_mh_no_high), aes(pm25, RR, color = sample)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2) +
  labs(
    title = "Sensitivity to extreme PM2.5 days: Mental health",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio"
  ) +
  theme_bw()
```




```r
ggplot(bind_rows(curve_sui_main, curve_sui_no_high), aes(pm25, RR, color = sample)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2) +
  labs(
    title = "Sensitivity to extreme PM2.5 days: Suicide",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio"
  ) +
  theme_bw()
```

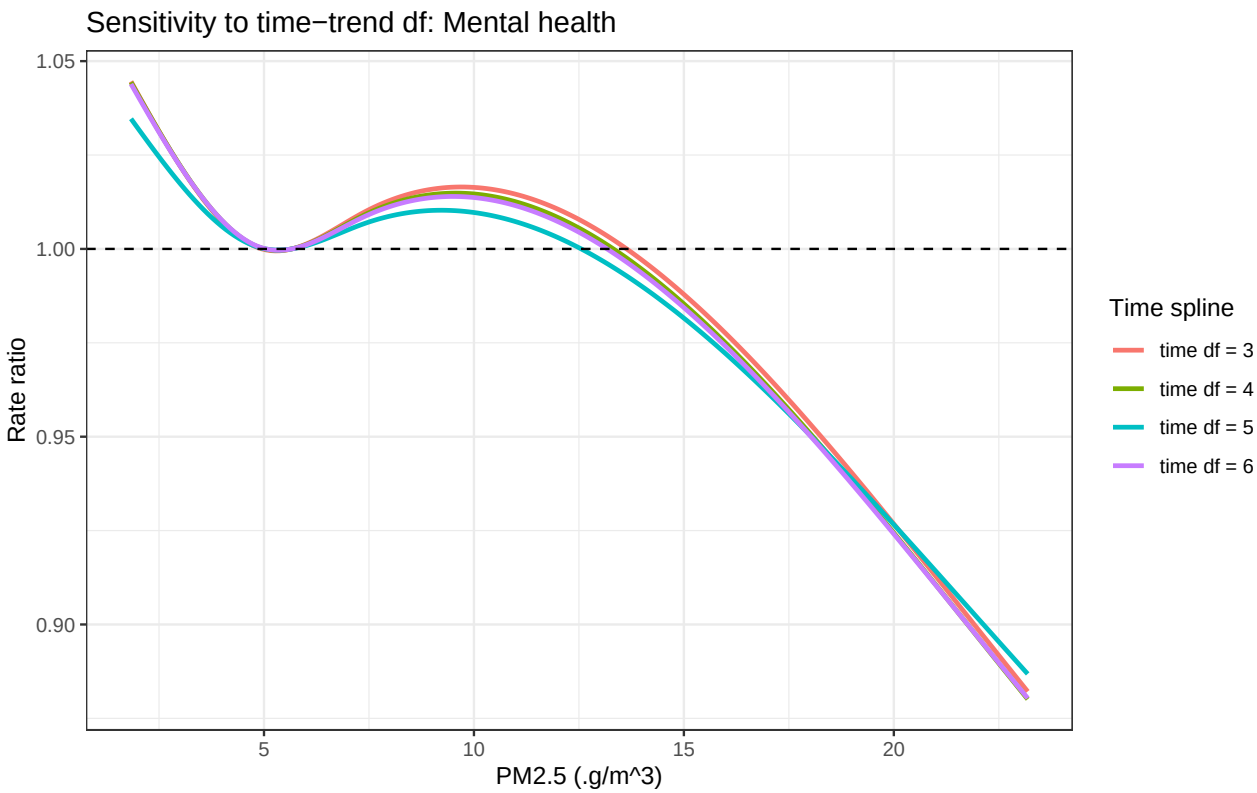


Sensitivity analysis 3: alternative time-trend df (3–6)

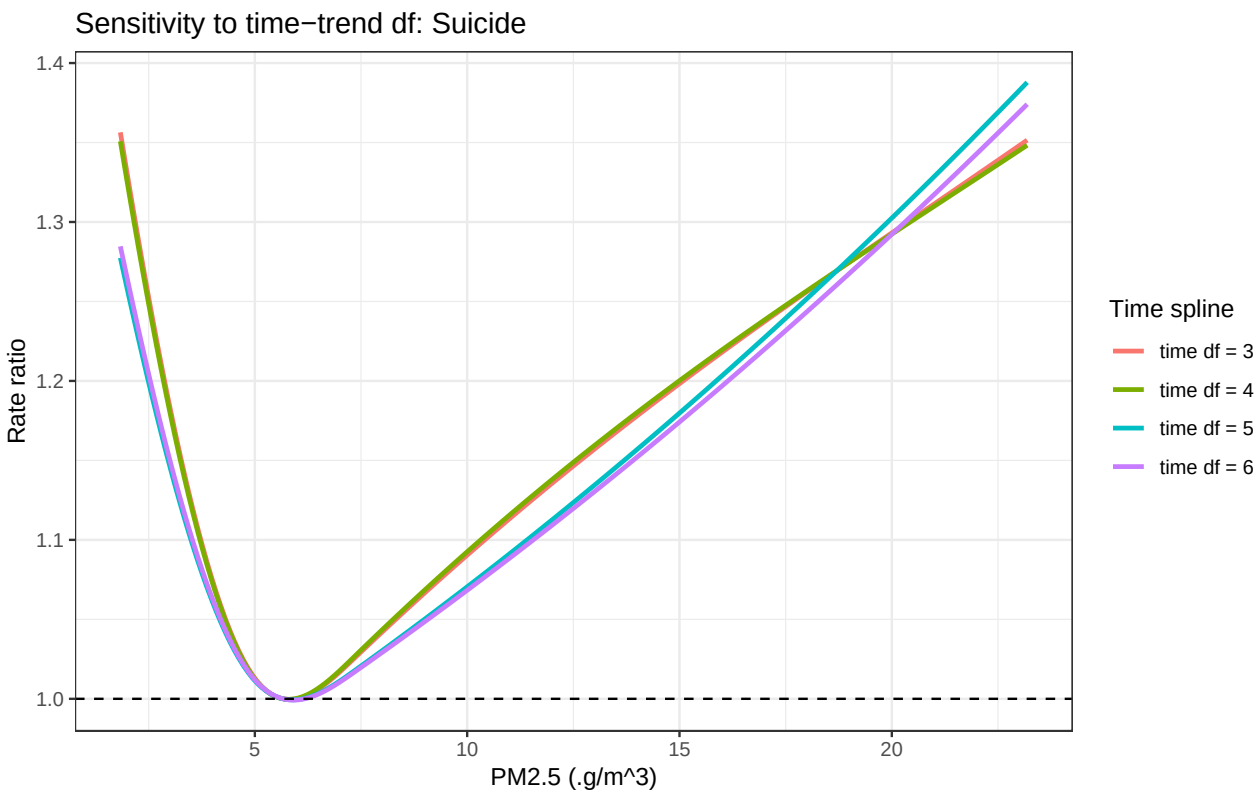
```
fit_time_sensitivity <- function(outcome, time_df, data) {
  form <- as.formula(
    paste0(outcome,
           " ~ ns(pm25, 3) + ns(time, ", time_df, ") + ns(tmean, 3) + is_weekend + precip")
  )

  glm(form, family = quasipoisson(link = "log"), data = data)
}

time_df_results_mh <- map_df(3:6, function(k) {
  mod <- fit_time_sensitivity("mental_health", k, df)
  curve <- make_er_curve(mod, df, "Mental health")
  curve$time_df <- paste0("time df = ", k)
  curve
})

time_df_results_sui <- map_df(3:6, function(k) {
  mod <- fit_time_sensitivity("suicide", k, df)
  curve <- make_er_curve(mod, df, "Suicide")
  curve$time_df <- paste0("time df = ", k)
  curve
})
```

```
ggplot(time_df_results_mh, aes(pm25, RR, color = time_df)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2) +
```

```r
  labs(
    title = "Sensitivity to time-trend df: Mental health",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio",
    color = "Time spline"
  ) +
  theme_bw()
```




```r
ggplot(time_df_results_sui, aes(pm25, RR, color = time_df)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2) +
  labs(
    title = "Sensitivity to time-trend df: Suicide",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio",
    color = "Time spline"
  ) +
  theme_bw()
```

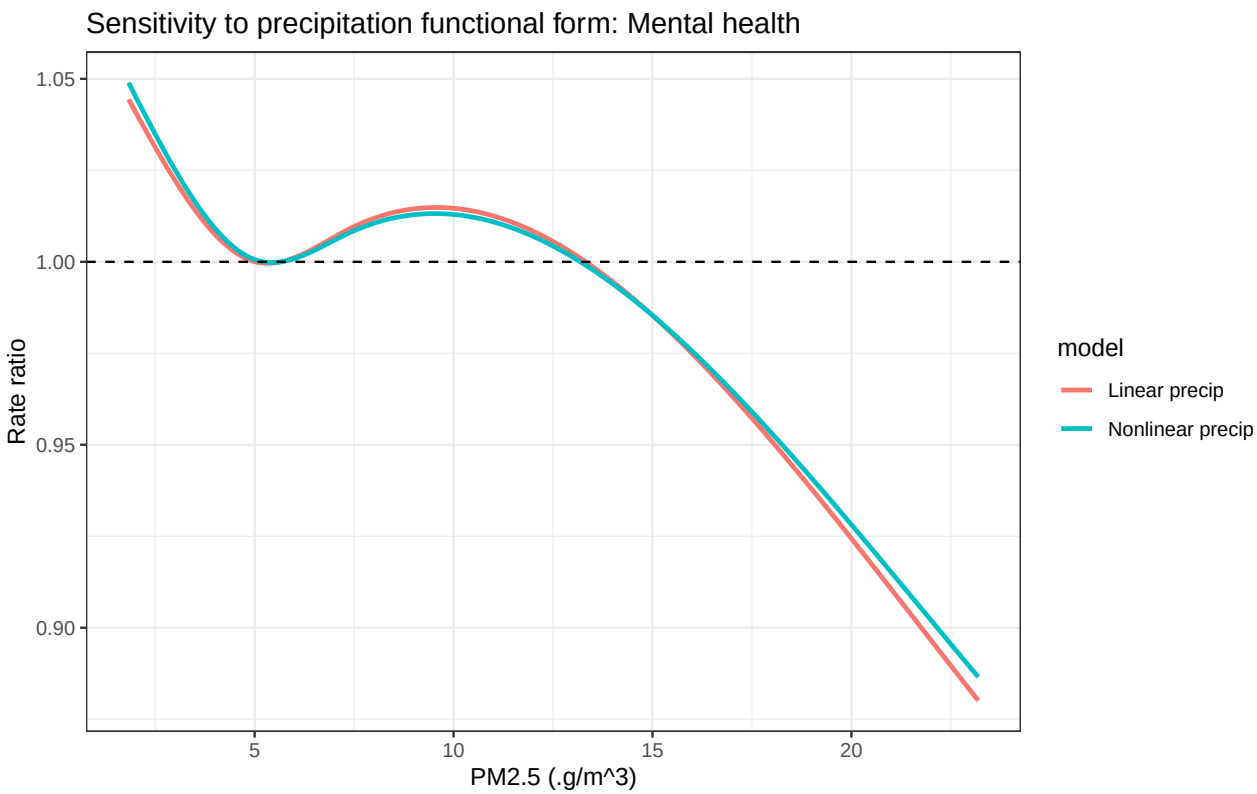


Sensitivity analysis 4: nonlinear precipitation

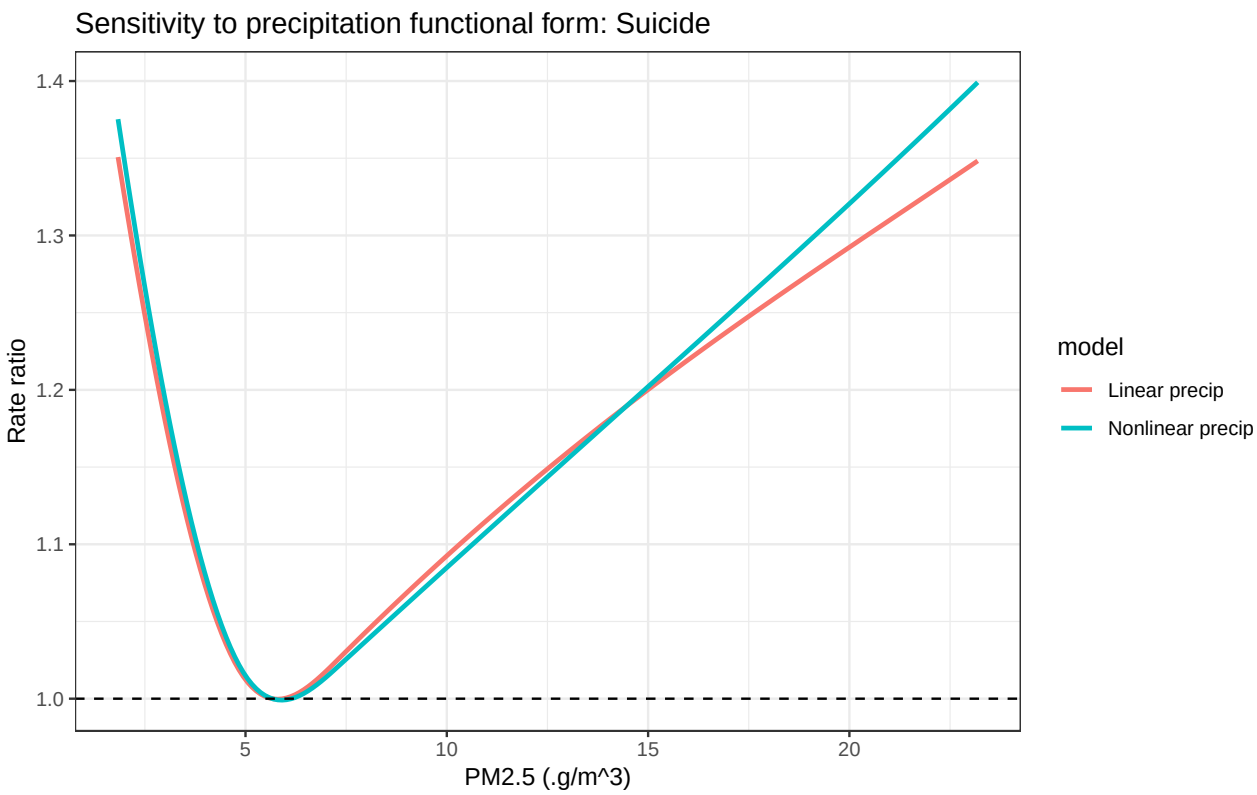
```
fit_mh_nl_precip <- glm(
  mental_health ~ ns(pm25, 3) + ns(time, 4) + ns(tmean, 3) + is_weekend + ns(precip, 3),
  family = quasipoisson(link = "log"),
  data = df
)

fit_sui_nl_precip <- glm(
  suicide ~ ns(pm25, 3) + ns(time, 4) + ns(tmean, 3) + is_weekend + ns(precip, 3),
  family = quasipoisson(link = "log"),
  data = df
)

curve_mh_nl_precip <- make_er_curve(fit_mh_nl_precip, df, "Mental health") |>
  mutate(model = "Nonlinear precip")

curve_sui_nl_precip <- make_er_curve(fit_sui_nl_precip, df, "Suicide") |>
  mutate(model = "Nonlinear precip")

curve_mh_main2 <- curve_mh |> mutate(model = "Linear precip")
curve_sui_main2 <- curve_sui |> mutate(model = "Linear precip")
```

```
ggplot(bind_rows(curve_mh_main2, curve_mh_nl_precip), aes(pm25, RR, color = model)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2) +
  labs(
    title = "Sensitivity to precipitation functional form: Mental health",
```

```
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio"
  ) +
  theme_bw()
```




```
ggplot(bind_rows(curve_sui_main2, curve_sui_nl_precip), aes(pm25, RR, color = model)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 1, linetype = 2) +
  labs(
    title = "Sensitivity to precipitation functional form: Suicide",
    x = "PM2.5 (\u03bcg/m^3)",
    y = "Rate ratio"
  ) +
  theme_bw()
```



Summary table

```
results_summary <- bind_rows(
  rr_mh_linear  |> mutate(outcome = "mental_health"),
  rr_sui_linear |> mutate(outcome = "suicide")
) |>
  select(outcome, RR_10, lower_95, upper_95)

results_summary
```

```
##                outcome     RR_10  lower_95 upper_95
## pm25...1 mental_health 0.9729817 0.9317521 1.016036
## pm25...2       suicide 1.0874212 0.8360522 1.414367
```

Quasi-Poisson Time Series GLM (Yiming Cao)

Model fitting

Using the same data loaded above. The main model keeps PM2.5 as a linear exposure term — the goal here is to estimate the rate ratio per 10 g/m^3 increase, adjusting for seasonal trend, temperature, weekend effect, and precipitation.

```
library(splines)

# time variable as days since Jan 1
df$time_days <- as.numeric(df$date - min(df$date))

# mental health model
```

```r
fit_qp_mh <- glm(mental_health ~ pm25 + ns(time_days, 4) + ns(tmean, 3) +
                    is_weekend + precip,
                  family = quasipoisson(link = "log"),
                  data = df)

# suicide model
fit_qp_sui <- glm(suicide ~ pm25 + ns(time_days, 4) + ns(tmean, 3) +
                     is_weekend + precip,
                   family = quasipoisson(link = "log"),
                   data = df)
```

```r
summary(fit_qp_mh)
```

```
##
## Call:
## glm(formula = mental_health ~ pm25 + ns(time_days, 4) + ns(tmean,
##     3) + is_weekend + precip, family = quasipoisson(link = "log"),
##     data = df)
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         6.1066967  0.0481988 126.698  < 2e-16 ***
## pm25               -0.0026775  0.0022112  -1.211  0.22845
## ns(time_days, 4)1 -0.1815532  0.0759208  -2.391  0.01844 *
## ns(time_days, 4)2 -0.1248743  0.0477825  -2.613  0.01019 *
## ns(time_days, 4)3 -0.0700981  0.0718951  -0.975  0.33164
## ns(time_days, 4)4 -0.0340114  0.0309432  -1.099  0.27404
## ns(tmean, 3)1       0.2261008  0.0482952   4.682 7.97e-06 ***
## ns(tmean, 3)2       0.4913475  0.1065191   4.613 1.05e-05 ***
## ns(tmean, 3)3       0.2444941  0.0726428   3.366  0.00104 **
## is_weekend         -0.1859232  0.0145624 -12.767  < 2e-16 ***
## precip             -0.0020202  0.0008409  -2.402  0.01792 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 2.238154)
##
##     Null deviance: 727.73  on 123  degrees of freedom
## Residual deviance: 254.69  on 113  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 3
```

```r
summary(fit_qp_sui)
```

```
##
## Call:
## glm(formula = suicide ~ pm25 + ns(time_days, 4) + ns(tmean, 3) +
##     is_weekend + precip, family = quasipoisson(link = "log"),
##     data = df)
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.734732   0.350062   2.099  0.03806 *
## pm25                0.008530   0.013405   0.636  0.52584
```

```
## ns(time_days, 4)1 -0.296227   0.468428  -0.632  0.52841
## ns(time_days, 4)2 -0.247127   0.297292  -0.831  0.40758
## ns(time_days, 4)3 -0.424140   0.464401  -0.913  0.36303
## ns(time_days, 4)4 -0.164994   0.205850  -0.802  0.42451
## ns(tmean, 3)1      0.967794   0.315754   3.065  0.00272 **
## ns(tmean, 3)2      2.222648   0.781310   2.845  0.00528 **
## ns(tmean, 3)3      0.508253   0.452109   1.124  0.26332
## is_weekend        -0.031116   0.088301  -0.352  0.72521
## precip            -0.004363   0.005364  -0.813  0.41775
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 0.9544157)
##
##     Null deviance: 129.88  on 123  degrees of freedom
## Residual deviance: 108.12  on 113  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```

Dispersion check

```r
cat("Mental health dispersion:", round(summary(fit_qp_mh)$dispersion, 2), "\n")
```

```
## Mental health dispersion: 2.24
```

```r
cat("Suicide dispersion:", round(summary(fit_qp_sui)$dispersion, 2), "\n")
```

```
## Suicide dispersion: 0.95
```

Rate ratios per 10 g/m³ PM2.5

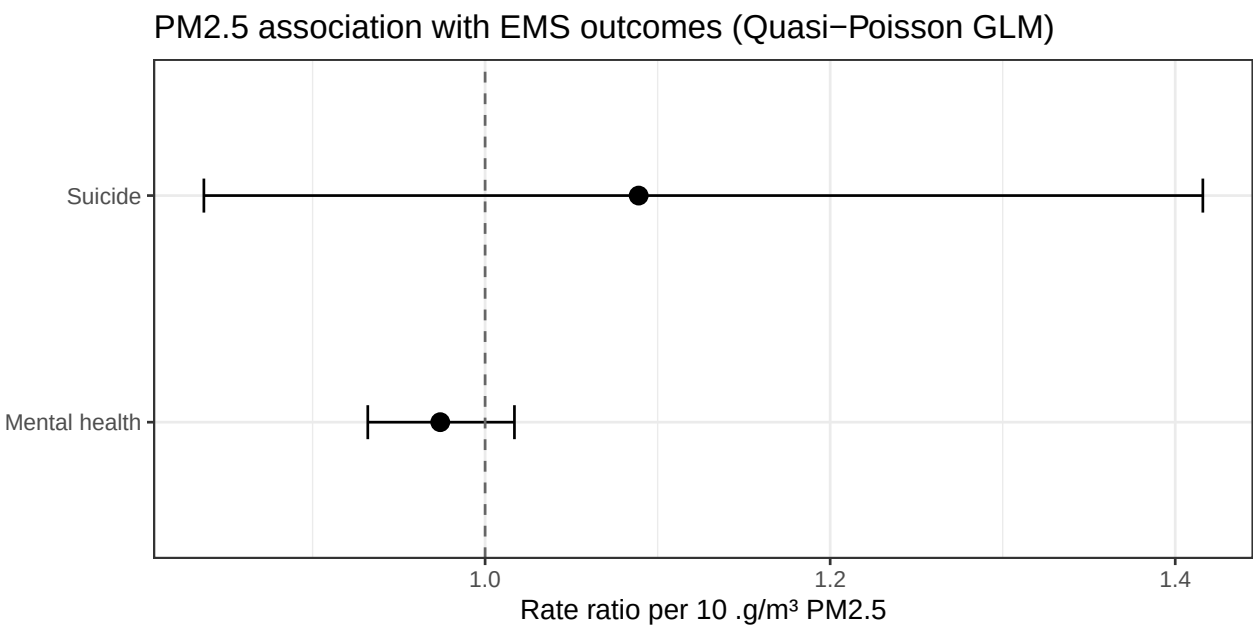
```r
extract_rr <- function(model, label) {
  b <- coef(model)["pm25"]
  se <- sqrt(vcov(model)["pm25", "pm25"])
  data.frame(
    outcome = label,
    RR = exp(b * 10),
    lower = exp((b - 1.96 * se) * 10),
    upper = exp((b + 1.96 * se) * 10),
    p = summary(model)$coefficients["pm25", 4]
  )
}

rr_table <- rbind(
  extract_rr(fit_qp_mh, "Mental health"),
  extract_rr(fit_qp_sui, "Suicide")
)
rr_table$RR <- round(rr_table$RR, 3)
rr_table$lower <- round(rr_table$lower, 3)
rr_table$upper <- round(rr_table$upper, 3)
rr_table$p <- round(rr_table$p, 3)
rr_table
```

```
##              outcome    RR lower upper     p
```

```
## pm25  Mental health 0.974 0.932 1.017 0.228
## pm251       Suicide 1.089 0.837 1.416 0.526
```

Rate ratio forest plot

```
ggplot(rr_table, aes(x = RR, y = outcome)) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.15) +
  geom_vline(xintercept = 1, linetype = "dashed", color = "gray40") +
  labs(x = "Rate ratio per 10  g/m³ PM2.5", y = NULL,
       title = "PM2.5 association with EMS outcomes (Quasi-Poisson GLM)") +
  theme_bw(base_size = 11)
```

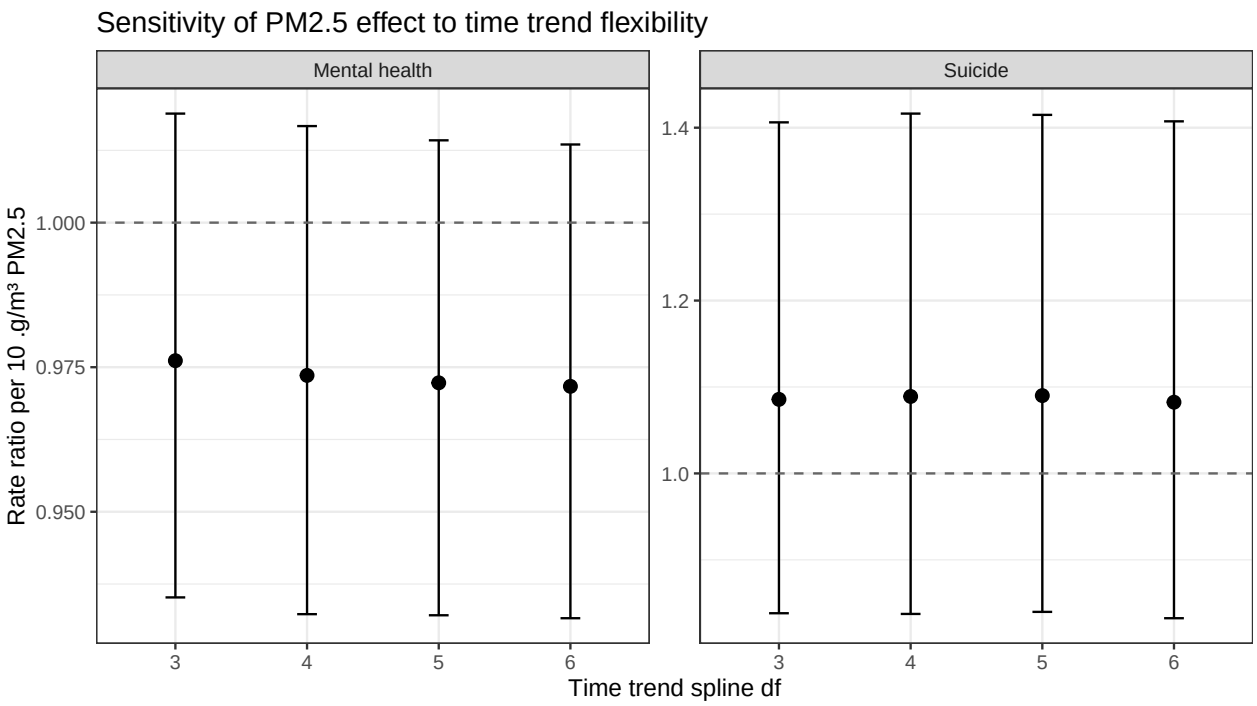


Sensitivity analysis: time trend df (3–6)

```
sens_df <- data.frame()
for (d in 3:6) {
  fit_tmp <- glm(mental_health ~ pm25 + ns(time_days, d) + ns(tmean, 3) +
                   is_weekend + precip,
                 family = quasipoisson, data = df)
  b <- coef(fit_tmp)["pm25"]
  se <- sqrt(vcov(fit_tmp)["pm25", "pm25"])

  sens_df <- rbind(sens_df, data.frame(
    df_time = d,
    outcome = "Mental health",
    RR = exp(b * 10),
    lower = exp((b - 1.96 * se) * 10),
    upper = exp((b + 1.96 * se) * 10)
  ))

  fit_tmp2 <- glm(suicide ~ pm25 + ns(time_days, d) + ns(tmean, 3) +
                    is_weekend + precip,
```

```
                    family = quasipoisson, data = df)
  b2 <- coef(fit_tmp2)["pm25"]
  se2 <- sqrt(vcov(fit_tmp2)["pm25", "pm25"])

  sens_df <- rbind(sens_df, data.frame(
    df_time = d,
    outcome = "Suicide",
    RR = exp(b2 * 10),
    lower = exp((b2 - 1.96 * se2) * 10),
    upper = exp((b2 + 1.96 * se2) * 10)
  ))
}
```

```
ggplot(sens_df, aes(x = factor(df_time), y = RR)) +
  geom_point(size = 2.5) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray40") +
  facet_wrap(~outcome, scales = "free_y") +
  labs(x = "Time trend spline df", y = "Rate ratio per 10  g/m³ PM2.5",
       title = "Sensitivity of PM2.5 effect to time trend flexibility") +
  theme_bw(base_size = 11)
```



Sensitivity analysis: exclude high PM2.5 days

```
pm95_cutoff <- quantile(df$pm25, 0.95)
df_trimmed <- df[df$pm25 <= pm95_cutoff, ]
cat("Excluded", nrow(df) - nrow(df_trimmed), "days with PM2.5 >",
    round(pm95_cutoff, 1), " g/m³\n")
```

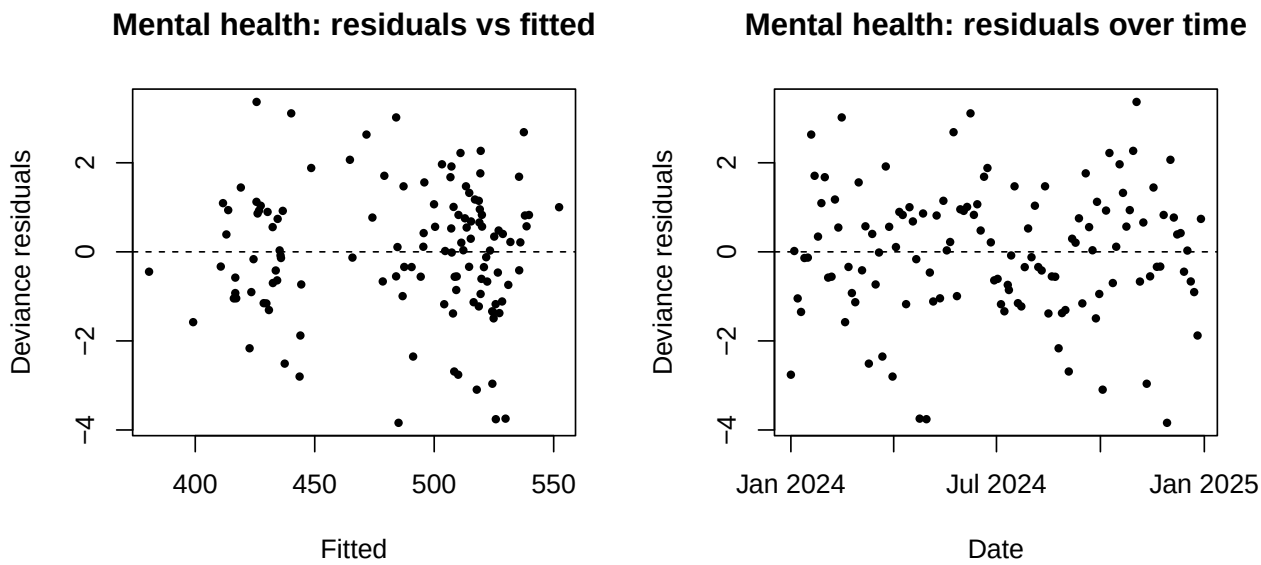
```
## Excluded 7 days with PM2.5 > 12.9  g/m³
```

```r
fit_trim_mh <- glm(mental_health ~ pm25 + ns(time_days, 4) + ns(tmean, 3) +
                      is_weekend + precip,
                    family = quasipoisson, data = df_trimmed)
fit_trim_sui <- glm(suicide ~ pm25 + ns(time_days, 4) + ns(tmean, 3) +
                       is_weekend + precip,
                     family = quasipoisson, data = df_trimmed)

rr_trim <- rbind(
  extract_rr(fit_trim_mh, "Mental health (trimmed)"),
  extract_rr(fit_trim_sui, "Suicide (trimmed)")
)
rr_trim$RR <- round(rr_trim$RR, 3)
rr_trim$lower <- round(rr_trim$lower, 3)
rr_trim$upper <- round(rr_trim$upper, 3)
rr_trim$p <- round(rr_trim$p, 3)

rbind(rr_table, rr_trim)
```

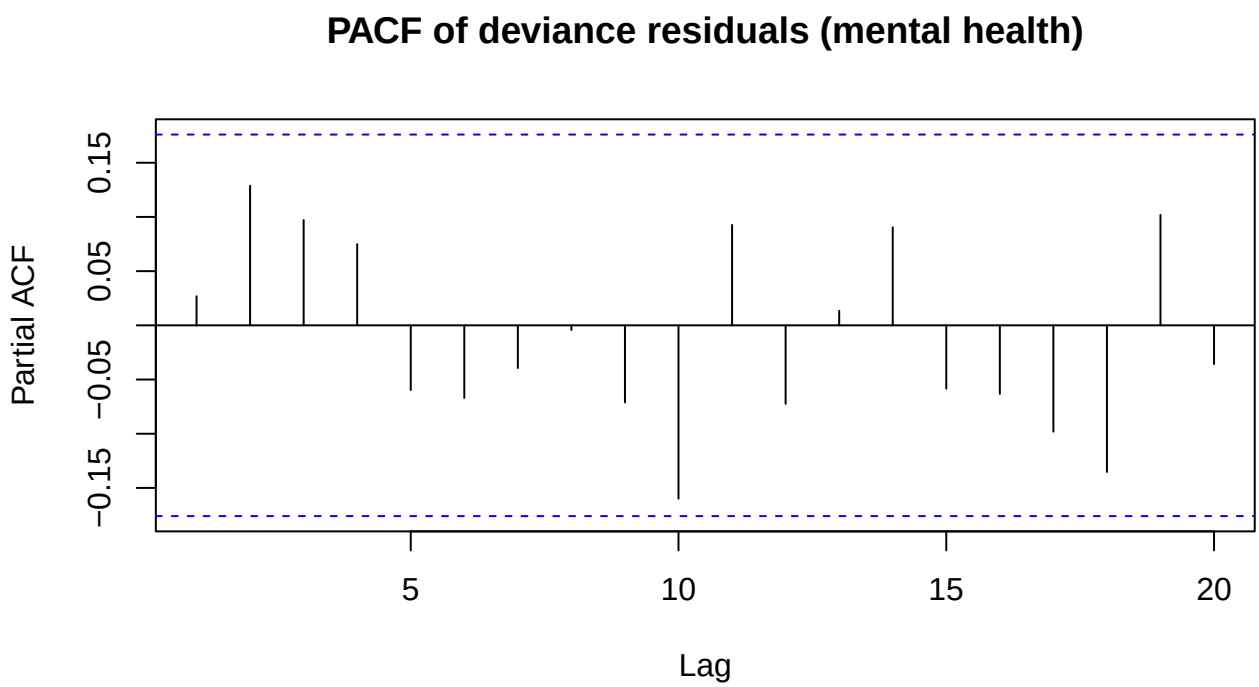
```
##                        outcome    RR lower upper     p
## pm25             Mental health 0.974 0.932 1.017 0.228
## pm251                  Suicide 1.089 0.837 1.416 0.526
## pm252  Mental health (trimmed) 1.015 0.956 1.077 0.629
## pm2511       Suicide (trimmed) 1.023 0.699 1.498 0.908
```

Residual diagnostics

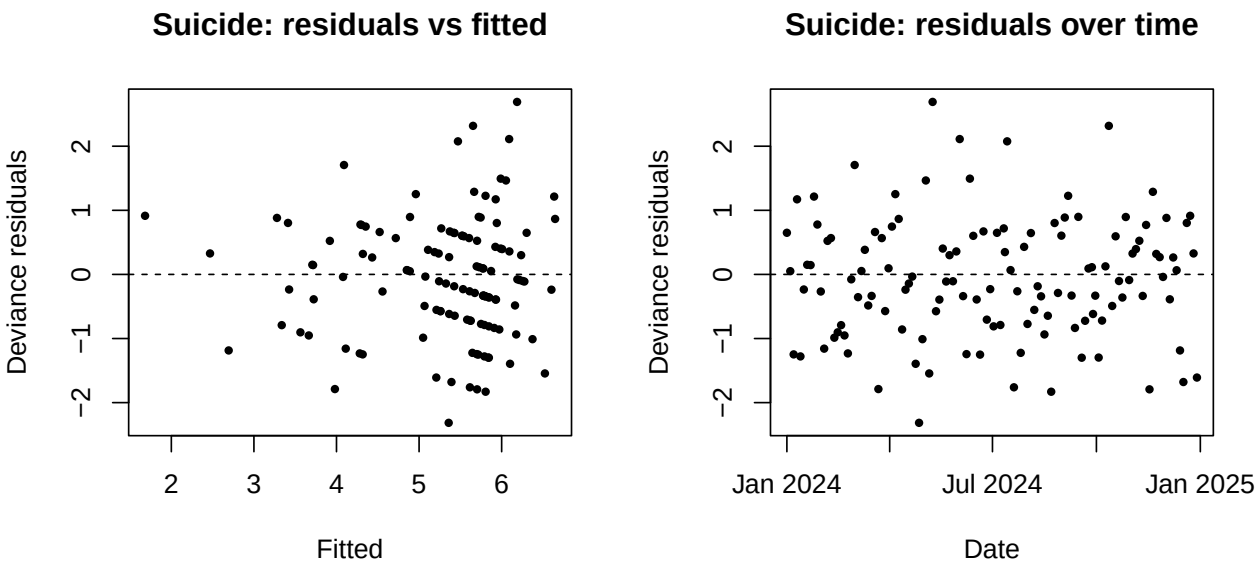
```r
par(mfrow = c(1, 2))
plot(fitted(fit_qp_mh), residuals(fit_qp_mh, type = "deviance"),
     xlab = "Fitted", ylab = "Deviance residuals",
     main = "Mental health: residuals vs fitted", pch = 16, cex = 0.7)
abline(h = 0, lty = 2)

plot(df$date, residuals(fit_qp_mh, type = "deviance"),
     xlab = "Date", ylab = "Deviance residuals",
     main = "Mental health: residuals over time", pch = 16, cex = 0.7)
abline(h = 0, lty = 2)
```




```
pacf(residuals(fit_qp_mh, type = "deviance"),
     main = "PACF of deviance residuals (mental health)")
```

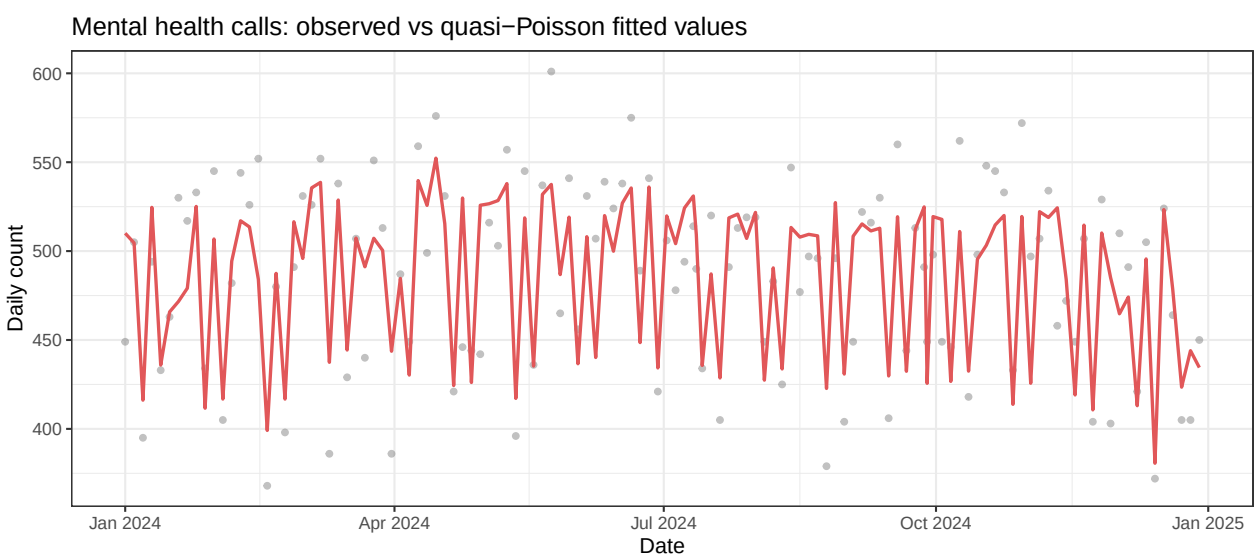



```
par(mfrow = c(1, 2))
plot(fitted(fit_qp_sui), residuals(fit_qp_sui, type = "deviance"),
     xlab = "Fitted", ylab = "Deviance residuals",
     main = "Suicide: residuals vs fitted", pch = 16, cex = 0.7)
abline(h = 0, lty = 2)

plot(df$date, residuals(fit_qp_sui, type = "deviance"),
     xlab = "Date", ylab = "Deviance residuals",
     main = "Suicide: residuals over time", pch = 16, cex = 0.7)
abline(h = 0, lty = 2)
```



Predicted vs observed

```
df$pred_mh <- fitted(fit_qp_mh)

ggplot(df, aes(x = date)) +
  geom_point(aes(y = mental_health), alpha = 0.4, size = 1.2, color = "gray40") +
  geom_line(aes(y = pred_mh), color = "#E15759", linewidth = 0.8) +
  labs(x = "Date", y = "Daily count",
       title = "Mental health calls: observed vs quasi-Poisson fitted values") +
  theme_bw(base_size = 11)
```



Results section outline (Quasi-Poisson GLM)

The quasi-Poisson GLM estimated a rate ratio of 0.974 (95% CI: 0.932–1.017) per 10 g/m³ increase in same-day PM2.5 for mental health EMS calls, after adjusting for seasonal trend, temperature, weekend, and precipitation ($p = 0.228$). For suicide-related calls, the estimated rate ratio was 1.089 (95% CI: 0.837–1.416, $p = 0.526$). Neither association was statistically significant.

Key points for the results write-up:

- The dispersion parameter for mental health was 2.24, confirming overdispersion relative to standard Poisson. For suicide calls the dispersion was 0.95, close to 1.
- The PM2.5 coefficient was not statistically significant for either outcome.
- Weekend indicator was a strong predictor for mental health calls (lower on weekends, $p < 0.001$), consistent with the EDA.
- Sensitivity analysis: the PM2.5 rate ratio for mental health was stable across time trend df = 3–6 (ranging from 0.972 to 0.976). However, excluding high-PM2.5 days (>95th percentile) shifted the estimate from 0.974 to 1.015, suggesting the slight negative association in the main model may be driven by a few high-exposure days.
- Residual diagnostics: check PACF plot for any remaining temporal autocorrelation after the time spline adjustment.